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Thermodynamics

NEET Chemistry · Previous Year Questions with Answer Key

103	45	1990 – 2026	English
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MCQ (Single Correct Answer)	102 questions
MCQ (More than One Correct Answer)	1 question
Answer key	at the end

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MCQ (Single Correct Answer) 102 questions

Q1. RE-NEET 2026

Open solution

Two moles of an ideal gas undergo free expansion from 10 L to 100 L at 300 K . The values of ΔS_{system} and $\Delta S_{surroundings}$ are

(R is universal gas constant)

(A) $\Delta S_{system} = 4.606R; \Delta S_{surroundings} = 0$

(B) $\Delta S_{system} = 0; \Delta S_{surroundings} = 0$

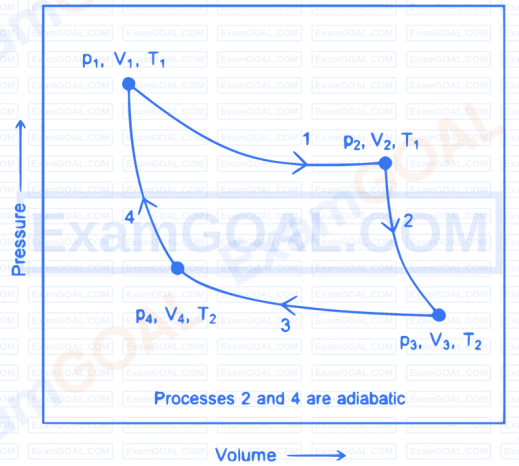
(C) $\Delta S_{system} = 4.606R; \Delta S_{surroundings} = -4.606R$

(D) $\Delta S_{system} = 0; \Delta S_{surroundings} = 4.606R$

Q2. RE-NEET 2026

Open solution

Consider the reversible processes for $1.0\ mol$ of an ideal gas as shown in the figure.



w_1, w_2, w_3 and w_4 represent work done (in calories) in the processes $1, 2, 3$ and 4 , respectively; ΔU_2 and ΔU_4 are changes in the internal energy for the processes 2 and 4, respectively.

[use $R = 2 cal K^{-1} mol^{-1}$]

The correct option is

(A) $w_1 + w_2 + w_3 + w_4 = 0$

(B) $w_1 + w_3 = -2\,T_1 \ln \frac{V_2}{V_1} - 2\,T_2 \ln \frac{V_4}{V_3}$

(C) $w_2 + w_4 = \Delta U_2 - \Delta U_4$

(D) $w_1 + w_2 = 2\,T_1 \ln \frac{V_2}{V_1}$

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Q3. RE-NEET 2026

A protein undergoes reversible thermal denaturation from its initial state $\mathbf{N}$ to denatured state $\mathbf{D}$ according to $\mathbf{N} \rightleftharpoons \mathbf{D}$. At $60^{\circ}\mathrm{C}$, the concentrations of both $\mathbf{N}$ and $\mathbf{D}$ are equal at equilibrium, and the standard enthalpy change of denaturation is $666\ \mathrm{kJ\ mol^{-1}}$. The standard entropy change $\Delta \mathrm{S}^{\circ}$ in $\mathrm{kJK^{-1}\ mol^{-1}}$) of the protein upon denaturation at $60^{\circ}\mathrm{C}$ is closest to

(A) 11.1 (B) 2.0 (C) 2000.0 (D) 333.0

Open solution

Q4. NEET 2026

Consider the following reaction :

$2\ \mathrm{A(\ g)} + \mathrm{B(\ g)} \rightarrow 2\mathrm{D(\ g)}$

$\Delta \mathrm{U}^{\ominus} = -10\ \mathrm{kJ\ mol^{-1}}$ and $\Delta \mathrm{S}^{\ominus} = -44\mathrm{JK^{-1}}$ at $298\ \mathrm{K}$.

Identify the correct option with $\Delta \mathrm{G}^{\ominus}$ for the reaction and spontaneity of the reaction at 298 K .

(Given : $\mathrm{R} = 8.31\ \mathrm{J\ mol^{-1}\ K^{-1}}$)

(A) $-1.635\ \mathrm{kJ\ mol^{-1}}$, spontaneous (B) $-0.63568\ \mathrm{kJ\ mol^{-1}}$, spontaneous

(C) $+0.63568\ \mathrm{kJ\ mol^{-1}}$, non-spontaneous (D) $+1.635\ \mathrm{kJ\ mol^{-1}}$, non-spontaneous

Open solution

Q5. NEET 2026

At a certain temperature, $\mathrm{T(K)}$, during a process, 500 J is absorbed by the system and work of 200 J is done by the system. Then change in internal energy of the system is :

(A) 400 J (B) 300 J (C) 700 J (D) 500 J

Open solution

Q6. NEET 2025

The standard heat of formation, in $\mathrm{kcal/mol}$ of $\mathrm{Ba^{2+}}$ is : [Given : standard heat of formation of $\mathrm{SO_4^{2-}}$ ion $(\mathrm{aq}) = -216\mathrm{kcal/mol}$, standard heat of crystallisation of $\mathrm{BaSO_4(\ s)} = -4.5\mathrm{kcal/mol}$, standard heat of formation of $\mathrm{BaSO_4(\ s)} = -349\mathrm{kcal/mol}$]

(A) +133.0 (B) +220.5 (C) -128.5 (D) -133.0

Open solution

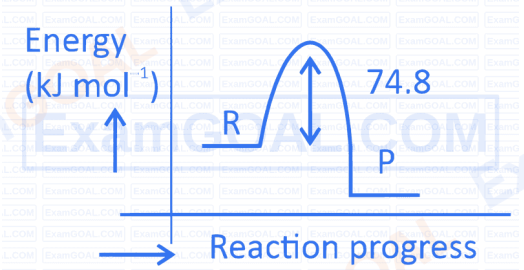
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Q7. NEET 2025

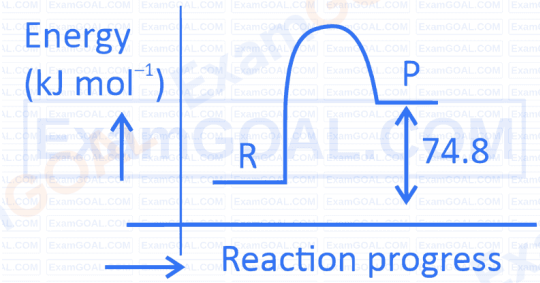
$C(s) + 2H_2(g) \rightarrow CH_4(g); \Delta H = -74.8\ kJ\ mol^{-1}$. Which of the following diagrams gives an accurate representation of the above reaction? [$R \rightarrow$ reactants; $P \rightarrow$ products]

Open solution

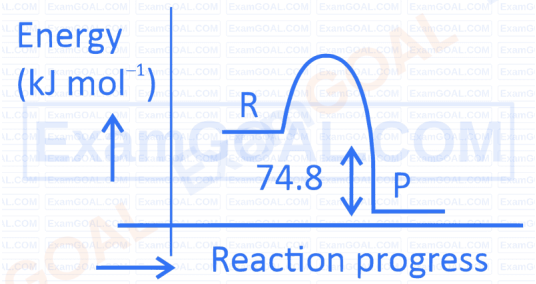
(A)



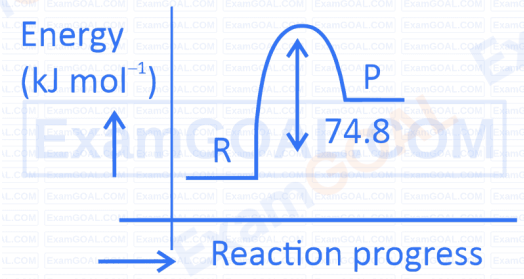
(B)



(C)



(D)



Q8. NEET 2024 (Re-Examination)

Choose the correct statement for the work done in the expansion and heat absorbed or released when 5 litres of an ideal gas at 10 atmospheric pressure isothermally expands into vacuum until volume is 15 litres :

Open solution

(A) Both the heat and work done will be greater than zero

(B) Heat absorbed will be less than zero and work done will be positive

(C) Work done will be zero and heat will also be zero

(D) Work done will be greater than zero and heat will remain zero

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Q9. NEET 2024 (Re-Examination)

For an endothermic reaction:

(A) q_p is negative.

(B) $\Delta_r H$ is positive.

(C) $\Delta_r H$ is negative.

(D) q_p is positive.

Choose the correct answer from the options given below:

(A) B and D **(B)** C and D **(C)** A and B **(D)** A and C

Open solution

Q10. NEET 2024 (Re-Examination)

For the following reaction at $300\ K$

$A_2(g) + 3B_2(g) \rightarrow 2AB_3(g)$

the enthalpy change is $+15\ kJ$, then the internal energy change is :

(A) $19988.4\ J$ **(B)** $200\ J$ **(C)** $1999\ J$ **(D)** $1.9988\ kJ$

Open solution

Q11. NEET 2024

In which of the following processes entropy increases?

A. A liquid evaporates to vapour.

B. Temperature of a crystalline solid lowered from $130\ K$ to $0\ K$.

C. $2NaHCO_{3(s)} \rightarrow Na_2CO_{3(s)} + CO_{2(g)} + H_2O_{(g)}$

D. $Cl_{2(g)} \rightarrow 2Cl_{(g)}$

Choose the correct answer from the options given below:

(A) A and C **(B)** A, B and D **(C)** A, C and D **(D)** C and D

Open solution

Q12. NEET 2024

Match List I with List II.

	List I (Process)		List II (Conditions)
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I **(B)** A-IV, B-II, C-III, D-I

(C) A-I, B-II, C-III, D-IV **(D)** A-II, B-III, C-IV, D-I

Open solution

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Q13. NEET 2024

The work done during reversible isothermal expansion of one mole of hydrogen gas at $25^{\circ}\mathrm{C}$ from pressure of 20 atmosphere to 10 atmosphere is (Given $\mathrm{R} = 2.0\ \mathrm{cal\ K^{-1}\ mol^{-1}}$)

(A) 0 calorie (B) -413.14 calories (C) 413.14 calories (D) 100 calories

Open solution

Q14. NEET 2023 Manipur

Consider the following reaction :-

$2\mathrm{H_2(\ g) + O_2(\ g) \rightarrow 2H_2O(g)}\Delta_{\mathrm{r}}\mathrm{H}^{\circ} = -483.64\ \mathrm{kJ}.$

What is the enthalpy change for decomposition of one mole of water? (Choose the right option).

(A) 120.9 kJ (B) 241.82 kJ (C) 18 kJ (D) 100 kJ

Open solution

Q15. NEET 2023

The equilibrium concentrations of the species in the reaction $\mathrm{A + B \rightleftharpoons C + D}$ are $2, 3, 10$ and $6\ \mathrm{mol\ L^{-1}}$, respectively at $300\ \mathrm{K}$. $\Delta\mathrm{G}^0$ for the reaction is $(\mathrm{R} = 2\mathrm{cal/mol\ K})$

(A) $-137.26\ \mathrm{cal}$ (B) $-1381.80\ \mathrm{cal}$ (C) $-13.73\ \mathrm{cal}$ (D) $1372.60\ \mathrm{cal}$

Open solution

Q16. NEET 2023

Which amongst the following options is the correct relation between change in enthalpy and change in internal energy?

(A) $\Delta\mathrm{H} = \Delta\mathrm{U} + \Delta\mathrm{n_gRT}$ (B) $\Delta\mathrm{H} - \Delta\mathrm{U} = -\Delta\mathrm{nRT}$

(C) $\Delta\mathrm{H} + \Delta\mathrm{U} = \Delta\mathrm{nR}$ (D) $\Delta\mathrm{H} = \Delta\mathrm{U} - \Delta\mathrm{n_gRT}$

Open solution

Q17. NEET 2022 Phase 2

One mole of an ideal gas at 300 K is expanded isothermally from 1 L to 10 L volume. ΔU for this process is :

(Use R = 8.314 J $\mathrm{k^{-1}}$ $\mathrm{mol^{-1}}$)

(A) 0 J (B) 1260 J (C) 2520 J (D) 5040 J

Open solution

Q18. NEET 2022 Phase 2

A vessel contains 3.2 g of dioxygen gas at STP (273.15 K and 1 atm pressure). The gas is now transferred to another vessel at constant temperature, where pressure becomes one third of the original pressure. The volume of new vessel in L is : (Given : molar volume at STP is 22.4 L)

(A) 67.2 (B) 6.72 (C) 2.24 (D) 22.4

Open solution

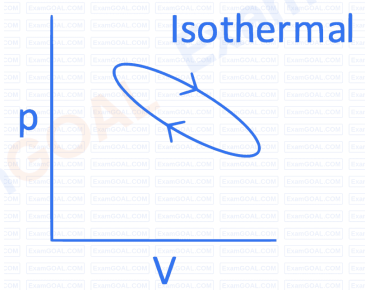
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Q19. NEET 2022 Phase 1

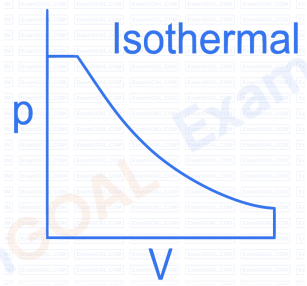
Which of the following p-V curve represents maximum work done?

Open solution

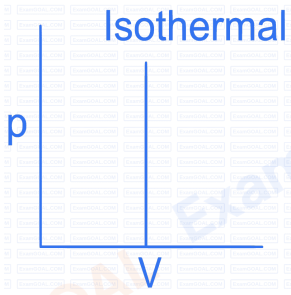
(A)



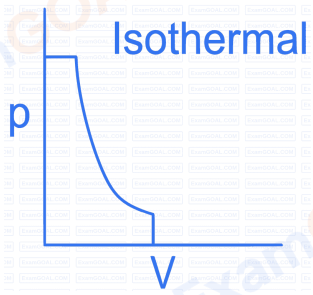
(B)



(C)



(D)



Q20. NEET 2021

Which one among the following is the correct option for right relationship between C_P and C_V for one mole of ideal gas?

Open solution

(A) $C_V = RC_P$ (B) $C_P + C_V = R$ (C) $C_P - C_V = R$ (D) $C_P = RC_V$

Q21. NEET 2021

For irreversible expansion of an ideal gas under isothermal condition, the correct option is :

Open solution

(A) $\Delta U \neq 0, \Delta S_{total} = 0$ (B) $\Delta U = 0, \Delta S_{total} = 0$

(C) $\Delta U \neq 0, \Delta S_{total} \neq 0$ (D) $\Delta U = 0, \Delta S_{total} \neq 0$

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Q22. NEET 2020 Phase 2

At standard conditions, if the change in the enthalpy for the following reaction is $-109\ \text{kJ mol}^{-1}$.

$\text{H}_{2(g)} + \text{Br}_{2(g)} \rightarrow 2\text{HBr}_{(g)}$

Given that bond energy of H_2 and Br_2 is $435\ \text{kJ mol}^{-1}$ and $192\ \text{kJ mol}^{-1}$, respectively, what is the bond energy (in kJ mol^{-1}) of HBr ?

(A) 259 (B) 368 (C) 736 (D) 518

Open solution

Q23. NEET 2020 Phase 2

If for a certain reaction $\Delta_r H$ is $30\ \text{kJ mol}^{-1}$ at $450\ \text{K}$, the value of $\Delta_r S$ (in $\text{J K}^{-1}\ \text{mol}^{-1}$) for which the same reaction will be spontaneous at the same temperature is

(A) -70 (B) 70 (C) -33 (D) 33

Open solution

Q24. NEET 2020 Phase 1

Hydrolysis of sucrose is given by the following reaction.
Sucrose + H_2O ⇌ Glucose + Fructose

If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of ΔrG^{Θ} at the same temperature will be :

(A) 8.314 J $mol^{-1}K^{-1} \times 300K \times \ln(2 \times 10^{13})$

(B) 8.314 J $mol^{-1}K^{-1} \times 300K \times \ln(3 \times 10^{13})$

(C) $-$ 8.314 J $mol^{-1}K^{-1} \times 300K \times \ln(4 \times 10^{13})$

(D) -8.314 J $mol^{-1}K^{-1} \times 300K \times \ln(2 \times 10^{13})$

Open solution

Q25. NEET 2020 Phase 1

For the reaction, 2Cl(g) $\rightarrow$ Cl_2(g), the correct option is :

(A) $\Delta_r H > 0$ and $\Delta_r S < 0$ (B) $\Delta_r H < 0$ and $\Delta_r S > 0$

(C) $\Delta_r H < 0$ and $\Delta_r S < 0$ (D) $\Delta_r H > 0$ and $\Delta_r S > 0$

Open solution

Q26. NEET 2020 Phase 1

The correct option for free expansion of an ideal gas under adiabatic condition is :

(A) q = 0, Δ T < 0 and w > 0 (B) q < 0, Δ T = 0 and w = 0

(C) q > 0, Δ T > 0 and w > 0 (D) q = 0, Δ T = 0 and w = 0

Open solution

Q27. NEET 2019 (Odisha)

An ideal gas expands isothermally from $10^{-3}\ \text{m}^3$ to $10^{-2}\ \text{m}^3$ at 300 K against a constant pressure of $10^5\ \text{N m}^{-2}$. The work done on the gas is

(A) $-900\ \text{kJ}$ (B) $+270\ \text{kJ}$ (C) $-900\ \text{J}$ (D) $+900\ \text{kJ}$

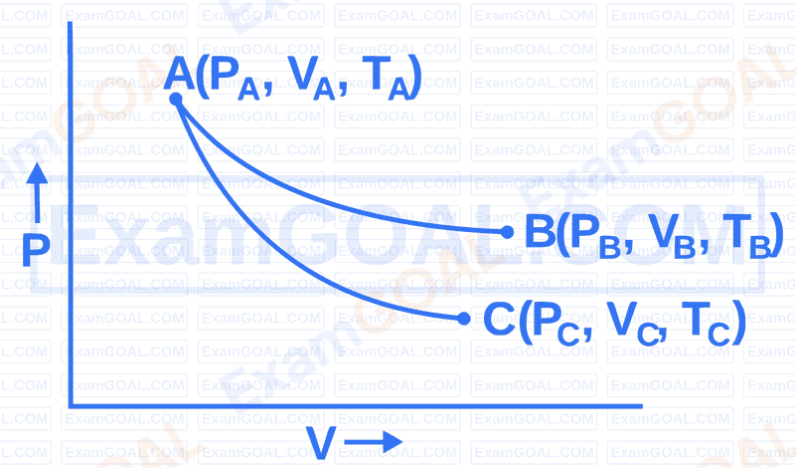
Open solution

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Q28. NEET 2019 (Odisha)

Open solution

Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.



AB $\rightarrow$ Isothermal expansion

AC $\rightarrow$ Adiabatic expansion

Which of the following options is **not** correct?

(A) $T_C > T_A$

(B) $\Delta S_{\text{isothermal}} > \Delta S_{\text{adiabatic}}$

(C) $T_A = T_B$

(D) $w_{\text{isothermal}} > w_{\text{adiabatic}}$

Q29. NEET 2019

Open solution

In which case change in entropy is negative?

(A) Sublimation of solid to gas

(B) $2H(g) \rightarrow H_2(g)$

(C) Evaporation of water

(D) Expansion of a gas at constant temperature

Q30. NEET 2019

Open solution

Under isothermal condition, a gas at 300 K expands from 0.1 L to 0.25 L against a constant external pressure of 2 bar. The work done by the gas is [Given that 1 L bar = 100 J]

(A) 25 J (B) 30 J (C) -30 J (D) 5 kJ

Q31. NEET 2018

Open solution

The bond dissociation energies of X_2 , Y_2 and XY are in the ratio of 1 : 0.5 : 1. ΔH for the formation of XY is –200 kJ mol^{-1}. The bond dissociation energy of X_2 will be

(A) 800 kJ mol^{-1} (B) 200 kJ mol^{-1} (C) 400 kJ mol^{-1} (D) 100 kJ mol^{-1}

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Q32. NEET 2017

For a given reaction, $\Delta H = 35.5\ kJ\ mol^{-1}$ and $\Delta S = 83.6\ J\ K^{-1}\ mol^{-1}$. The reaction is spontaneous at (Assume that ΔH and ΔS do not vary with temperature.)

(A) T > 425 K (B) all temperatures (C) T > 298 K (D) T < 425 K

Open solution

Q33. NEET 2017

A gas is allowed to expand in a well insulated container against a constant external pressure of 2.5 atm from an initial volume of 2.50 L to a final volume of 4.50 L. The change in internal energy ΔU of the gas in joules will be

(A) − 500 J (B) − 505 J (C) + 505 J (D) 1136.25 J

Open solution

Q34. NEET 2016 Phase 2

For a sample of perfect gas when its pressure is changed isothermally from p_i to p_f, the entropy change is given by

(A) $\Delta S = nR \ln\left(\frac{p_f}{p_i}\right)$ (B) $\Delta S = nR \ln\left(\frac{p_i}{p_f}\right)$

(C) $\Delta S = nRT \ln\left(\frac{p_f}{p_i}\right)$ (D) $\Delta S = RT \ln\left(\frac{p_i}{p_f}\right)$

Open solution

Q35. AIPMT 2015

The heat of combustion of carbon to CO_2 is −393.5 kJ/mol. The heat released upon formation of 35.2 g of CO_2 from carbon and oxygen gas is

(A) + 315 kJ (B) −630 kJ (C) − 3.15 kJ (D) − 315 kJ

Open solution

Q36. AIPMT 2014

Which of the following statements is correct for the spontaneous adsorption of a gas?

(A) ΔS is negative and, therefore ΔH should be highly positive.

(B) ΔS is negative and therefore, ΔH should be highly negative.

(C) ΔS is positive and therefore, ΔH should be negative .

(D) ΔS is positive and therefore, ΔH should also be highly positive.

Open solution

Q37. AIPMT 2014

For the reaction, $X_2O_{4(l)} \rightarrow 2XO_{2(g)}$

$\Delta U = 2.1$ kcal, $\Delta S = 20\ cal\ K^{-1}$ at 300 K

Hence, ΔG is

(A) 2.7 kcal (B) − 2.7 kcal (C) 9.3 kcal (D) − 9.3 kcal

Open solution

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Q38. NEET 2013 (Karnataka)

When 5 litres of a gas mixture of methane and propane is perfectly combusted at 0^oC and 1 atmosphere, 16 litres of oxygen at the same temperature and pressure is consumed. The amount of heat released from this combustion in kJ ($\Delta H_{comb.}$ (CH_4) = 890 kJ mol^{-1}, $\Delta H_{comb.}$ (C_3H_8) = 2220 kJ mol^{-1}) is

Open solution

(A) 38 (B) 317 (C) 477 (D) 32

Q39. NEET 2013 (Karnataka)

Three thermochemical equations are given below

(i) $C_{(graphite)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta_r H^o$ = x kJ mol^{-1}

(ii) $C_{(graphite)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$; $\Delta_r H^o$ = y kJ mol^{-1}

(iii) $CO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta_r H^o$ = z kJ mol^{-1}

Based on the above equations, find out which of the relationship given below is correct.

Open solution

(A) z = x + y (B) x = y + z (C) y = 2z – x (D) x = y – z

Q40. NEET 2013

A reaction having equal energies of activation for forward and reverse reactions has

Open solution

(A) $\Delta H = 0$ (B) $\Delta H = \Delta G = \Delta S = 0$

(C) $\Delta S = 0$ (D) $\Delta G = 0$

Q41. AIPMT 2012 Prelims

In which of the following reactions, standard reaction entropy change (ΔS^o) is positive and standard Gibb's energy change (ΔG^o) decreases sharply with increasing temperature ?

Open solution

(A) $C_{(graphite)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$

(B) $CO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}$

(C) $Mg_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow MgO_{(s)}$

(D) $\frac{1}{2}C_{(graphite)} + \frac{1}{2}O_{2(g)} \rightarrow \frac{1}{2}CO_{2(g)}$

Q42. AIPMT 2012 Prelims

The enthalpy of fusion of water is 1.435 kcal/mol. The molar entropy change for the melting of ice at 0^oC is

Open solution

(A) 10.52 cal/(mol K) (B) 21.04 cal/(mol K)

(C) 5.260 cal/(mol K) (D) 0.526 cal/(mol K)

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Q43. AIPMT 2012 Prelims

Standard enthalpy of vaporisation $\Delta_{vap}H^{o}$ for water at 100°C is 40.66 kJ mol^{-1}. The internal energy of vaporisation of water at 100°C (in kJ mol^{-1}) is

(A) +37.56 (B) −43.76 (C) + 43.76 (D) + 40.66

Open solution

Q44. AIPMT 2011 Mains

Consider the following processes :

	H (kJ/mol)
1/2A → B	+150
3B → 2C + D	-125
E + A → 2D	+350

For B + D → E + 2C, ΔH will be

(A) 525 kJ/mol (B) −175 kJ/mol (C) − 325 kJ/mol (D) 325 kJ/mol

Open solution

Q45. AIPMT 2011 Prelims

If the enthalpy change for the transition of liquid water to steam is 30 kJ mol^{-1} at 27°C, the entropy change for the process would be

(A) 10 J mol^{-1} K^{-1} (B) 1.0 J mol^{-1} K^{-1} (C) 0.1 J mol^{-1} K^{-1} (D) 100 J mol^{-1} K^{-1}

Open solution

Q46. AIPMT 2011 Prelims

Enthalpy change for the reaction,
$4H_{(g)} \rightarrow 2H_{2(g)}$ is −869.6 kJ
The dissociation energy of H − H bond is

(A) − 434.8 kJ (B) − 869.6 kJ (C) + 434.8 kJ (D) + 217.4 kJ

Open solution

Q47. AIPMT 2011 Prelims

Which of the following is correct option for free expansion of an ideal gas under adiabatic condition ?

(A) q = 0, $\Delta T \neq 0$, w = 0 (B) $q \neq 0$, $\Delta T = 0$, w = 0
(C) q = 0, $\Delta T = 0$, w = 0 (D) q = 0, $\Delta T < 0$, $w \neq 0$

Open solution

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Q48. AIPMT 2010 Mains

Match List I (Equations) with List II (Type of processes) and select the correct option.

Open solution

List I		List II	
Equations		Type of processes	
A.	$K_p > Q$	(i)	Non- spontaneous
B.	$\Delta G^o < RT \ln Q$	(ii)	Equilibrium
C.	$K_p = Q$	(iii)	Spontaneous and endothermic
D.	$T > \frac{\Delta H}{\Delta S}$	(iv)	Spontaneous

(A) A - (i), B - (ii), C - (iii), D - (iv)
(B) A - (iii), B - (iv), C - (ii), D - (i)
(C) A - (iv), B - (i), C - (ii), D - (iii)
(D) A - (ii), B - (i), C - (iv), D - (iii)

Q49. AIPMT 2010 Mains

Three moles of an ideal gas expanded spontaneously into vacuum. The work done will be

Open solution

(A) infinite **(B)** 3 Joules **(C)** 9 Joules **(D)** zero

Q50. AIPMT 2010 Mains

For vaporization of water at 1 atmospheric pressure, the values of ΔH and ΔS are 40.63 kJ mol^{-1} and 108.8 J K^{-1} mol^{-1}, respectively. The temperature when Gibb's energy change (ΔG) for this transformation will be zero, is

Open solution

(A) 273.4 K **(B)** 393.4 K **(C)** 373.4 K **(D)** 293.4 K

Q51. AIPMT 2010 Mains

The following two reactions are known

$Fe_2O_{3(s)} + 3CO_{(g)} \rightarrow 2Fe_{(s)} + 3CO_{2(g)}$; $\Delta H = -26.8$ kJ

$FeO_{(s)} + CO(g) \rightarrow Fe_{(s)} + CO_{2(g)}$; $\Delta H = -16.5$ kJ

The value of ΔH for the following reaction

$Fe_2O_{3(s)} + CO_{(g)} \rightarrow 2FeO_{(s)} + CO_{2(g)}$ is

Open solution

(A) + 10.3 kJ **(B)** − 43.3 kJ **(C)** − 10.3 kJ **(D)** + 6.2 kJ

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Q52. AIPMT 2010 Prelims

For an endothermic reaction, energy of activation is E_a and enthalpy of reaction is ΔH (both of these in kJ/mol). Minimum value of E_a will be

(A) less than ΔH (B) equal to ΔH (C) more than ΔH (D) equal to zero

Open solution

Q53. AIPMT 2010 Prelims

Standard entropies of X_2, Y_2 and XY_3 are 60, 40 and 50 J K^{-1} mol^{-1} respectively. For the reaction

$1/2X_2 + 3/2Y_2 \rightleftharpoons XY_3$, $\Delta H = -30$ kJ,

to be at equilibrium, the temperature should be

(A) 750 K (B) 1000 K (C) 1250 K (D) 500 K

Open solution

Q54. AIPMT 2009

The values of ΔH and ΔS for the reaction,

$C_{(graphite)} + CO_2(g) \rightarrow 2CO_{(g)}$

are 170 kJ and 170 J K^{-1}, respectively. This reaction will be spontaneous at

(A) 910 K (B) 1110 K (C) 510 K (D) 710 K

Open solution

Q55. AIPMT 2009

From the following bond energies :

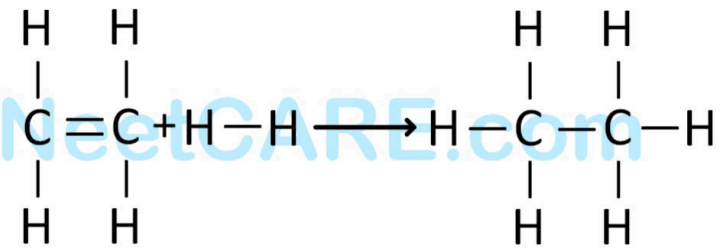
H – H bond energy : 431.37 kJ mol^{-1}

C = C bond energy : 606.10 kJ mol^{-1}

C – C bond energy : 336.49 kJ mol^{-1}

C – H bond energy : 410.50 kJ mol^{-1}

Enthalpy for the reaction,



will be

(A) – 243.6 kJ mol^{-1} (B) – 120.0 kJ mol^{-1} (C) 553.0 kJ mol^{-1} (D) 1523.6 kJ mol^{-1}

Open solution

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Q56. AIPMT 2008

Bond dissociation enthalpy of H_2, Cl_2 and HCl are 434, 242 and 431 kJ mol^{-1} respectively. Enthalpy of formation of HCl is

(A) $-$ 93 kJ mol^{-1} (B) 245 kJ mol^{-1} (C) 93 kJ mol^{-1} (D) $-$ 245 kJ mol^{-1}

Open solution

Q57. AIPMT 2008

For the gas phase reaction,

$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

which of the following conditions are correct ?

(A) $\Delta H < 0$ and $\Delta S < 0$ (B) $\Delta H > 0$ and $\Delta S < 0$

(C) $\Delta H = 0$ and $\Delta S < 0$ (D) $\Delta H > 0$ and $\Delta S > 0$

Open solution

Q58. AIPMT 2008

Which of the following are not state functions ?

(I) q + w (II) q

(III) w (IV) H $-$ TS

(A) (I), (II) and (III) (B) (II) and (III)

(C) (I) and (IV) (D) (II), (III) and (IV)

Open solution

Q59. AIPMT 2007

Consider the following reactions :

(i) $H^+_{(aq)} + OH^-_{(aq)} = H_2O_{(l)}$, $\Delta H = -X_1$ kJ mol^{-1}

(ii) $H_{2(g)} + 1/2O_{2(g)} = H_2O_{(l)}$, $\Delta H = -X_2$ kJ mol^{-1}

(iii) $CO_{2(g)} + H_{2(g)} = CO_{(g)} + H_2O_{(l)}$, $\Delta H = -X_3$ kJ mol^{-1}

(iv) $C_2H_{2(g)} + 5/2O_{2(g)} = 2CO_{2(g)} + H_2O_{(l)}$, $\Delta H = +X_4$ kJ mol^{-1}

Enthalpy of formation of $H_2O_{(l)}$ is

(A) $+X_3$ kJ mol^{-1} (B) $-X_4$ kJ mol^{-1} (C) $+X_1$ kJ mol^{-1} (D) $-X_2$ kJ mol^{-1}

Open solution

Q60. AIPMT 2007

Given that bond energies of H $-$ H and Cl $-$ Cl are 430 kJ mol^{-1} and 240 kJ mol^{-1} respectively and ΔH_f for HCl is $-$ 90 kJ mol^{-1}, bond enthalpy of HCl is

(A) 380 kJ mol^{-1} (B) 425 kJ mol^{-1} (C) 245 kJ mol^{-1} (D) 290 kJ mol^{-1}

Open solution

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Q61. AIPMT 2006

Identify the correct statement for change of Gibb's energy for a system (ΔG_{system}) at constant temperature and pressure.

(A) If $\Delta G_{system} < 0$, the process is not spontaneous.

(B) If $\Delta G_{system} > 0$, the process is spontaneous.

(C) If $\Delta G_{system} = 0$, the system has attained equilibrium.

(D) If $\Delta G_{system} = 0$, the system is till moving in a particular direction.

Open solution

Q62. AIPMT 2006

Assume each reaction is carried out in an open container. For which reaction will $\Delta H = \Delta E$?

(A) $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$

(B) $H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$

(C) $C_{(s)} + 2H_2O_{(g)} \rightarrow 2H_{2(g)} + CO_{2(g)}$

(D) $PCl_{5(g)} \rightarrow PCl_{3(g)} + Cl_{2(g)}$

Open solution

Q63. AIPMT 2006

The enthalpy and entropy change for the reaction:
$Br_{2(l)} + Cl_{2(g)} \rightarrow 2BrCl_{(g)}$
are 30 kJ mol^{-1} and 105 J K^{-1} mol^{-1} respectively.
The temperature at which the reaction will be in equilibrium is

(A) 300 K (B) 285.7 K (C) 273 K (D) 450 K

Open solution

Q64. AIPMT 2006

The enthalpy of combustion of H_2, cyclohexene (C_6H_{10}) and cyclohexane (C_6H_{12}) are -241, -3800 and -3920 kJ per mol respectively. Heat of hydrogenation of cyclohexene is

(A) -121 kJ per mol (B) $+121$ kJ per mol (C) $+484$ kJ per mol (D) -484 kJ per mol

Open solution

Q65. AIPMT 2005

Which of the following pairs of a chemical reaction is certain to result in a spontaneous reaction ?

(A) Exothermic and increasing disorder

(B) Exothermic and decreasing disorder

(C) Endothermic and increasing disorder

(D) Endothermic and decreasing disorder

Open solution

Q66. AIPMT 2005

A reaction occurs spontaneously if

(A) $T\Delta S < \Delta H$ and both ΔH and ΔS are +ve

(B) $T\Delta S > \Delta H$ and ΔH is +ve and ΔS are $-$ve

(C) $T\Delta S > \Delta H$ and both ΔH and ΔS are +ve

(D) $T\Delta S = \Delta H$ and both ΔH and ΔS are +ve

Open solution

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Q67. AIPMT 2005

The absolute enthalpy of neutralisation of the reaction :

$MgO_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_2O_{(l)}$ will be

(A) -57.33 kJ mol^{-1}
(B) greater than -57.33 kJ mol^{-1}
(C) less than -57.33 kJ mol^{-1}
(D) 57.33 kJ mol^{-1}

Open solution

Q68. AIPMT 2004

If the bond energies of H $-$ H, Br $-$ Br, and H $-$ Br are 433, 192 and 364 kJ mol^{-1} respectively, the ΔH^o for the reaction

$H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$ is

(A) $-$ 261 kJ
(B) +103 kJ
(C) +261 kJ
(D) $-$103 kJ

Open solution

Q69. AIPMT 2004

Standard enthalpy and standard entropy changes for the oxidation of ammonia at 298 K are $-$ 382.64 kJ mol^{-1} and $-$ 145.6 J K^{-1} mol^{-1}, respectively. Standard Gibb's energy change for the same reaction at 298 K is

(A) $-$ 221.1 kJ mol^{-1}
(B) $-$339.3 kJ mol^{-1}
(C) $-$ 439.3 kJ mol^{-1}
(D) $-$ 523.2 kJ mol^{-1}

Open solution

Q70. AIPMT 2004

Considering entropy (S) as a thermodynamic parameter, the criterion for the spontaneity of any process is

(A) $\Delta S_{system} + \Delta S_{surroundings} > 0$
(B) $\Delta S_{system} - \Delta S_{surroundings} > 0$
(C) $\Delta S_{system} > 0$ only
(D) $\Delta S_{surroundings} > 0$ only

Open solution

Q71. AIPMT 2004

The work done during the expansion of a gas from a volume of 4 dm^3 to 6 dm^3 against a constant external pressure of 3 atm is (1 L atm = 101.32 J)

(A) $-$ 6 J
(B) $-$ 608 J
(C) + 304 J
(D) $-$ 304 J

Open solution

Q72. AIPMT 2003

For the reaction,
$C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_2(g) + 4H_2O_{(l)}$
at constant temperature, $\Delta H - \Delta E$ is

(A) + RT
(B) $-$3RT
(C) + 3RT
(D) $-$ RT

Open solution

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Q73. AIPMT 2003

The densities of graphite and diamond at 298 K are 2.25 and 3.31 g cm^{-3}, respectively. If the standard free energy difference (ΔG^o) is equal to 1895 J mol^{-1}, the pressure at which graphite will be transformed into diamond at 298 K is

Open solution

(A) 11.14×10^8 Pa
(B) 11.14×10^7 Pa
(C) 11.14×10^6 Pa
(D) 11.14×10^5 Pa

Q74. AIPMT 2003

What is the entropy change (in J K^{-1} mol^{-1}) when one mole of ice is converted into water at 0^oC? (The enthalpy change for the conversion of ice to liquid water is 6.0 kJ mol^{-1} at 0^oC).

Open solution

(A) 20.13
(B) 2.013
(C) 2.198
(D) 21.98

Q75. AIPMT 2003

Formation of solution from two components can be considered as
(i) Pure solvent $\rightarrow$ separated solvent molecules, ΔH_1
(ii) Pure solute $\rightarrow$ separated solute molecules, ΔH_2
(iii) Separated solvent and solute molecules $\rightarrow$ solution, ΔH_3
Solution so formed will be ideal if

Open solution

(A) $\Delta H_{soln} = \Delta H_1 + \Delta H_2 + \Delta H_3$
(B) $\Delta H_{soln} = \Delta H_1 + \Delta H_2 - \Delta H_3$
(C) $\Delta H_{soln} = \Delta H_1 - \Delta H_2 - \Delta H_3$
(D) $\Delta H_{soln} = \Delta H_3 - \Delta H_1 - \Delta H_2$

Q76. AIPMT 2003

For which one of the following equations is ΔH^o_{react} equal to ΔH^o_f for the product ?

Open solution

(A) $Xe_{(g)} + 2F_{2(g)} \rightarrow XeF_{4(g)}$
(B) $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)}$
(C) $N_{2(g)} + O_{3(g)} \rightarrow N_2O_{3(g)}$
(D) $CH_{4(g)} + 2Cl_{2(g)} \rightarrow CH_2Cl_{2(l)} + 2HCl_{(g)}$

Q77. AIPMT 2003

The molar heat capacity of water at constant pressure, C, is 75 J K^{-1} mol^{-1}. When 1.0 kJ of heat is supplied to 100 g of water which is free to expand, the increase in temperature of water is

Open solution

(A) 1.2 K
(B) 2.4 K
(C) 4.8 K
(D) 6.6 K

Q78. AIPMT 2002

Unit of entropy is

Open solution

(A) J K^{-1} mol^{-1}
(B) J mol^{-1}
(C) J^{-1} K^{-1} mol^{-1}
(D) J K mol^{-1}

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Q79. AIPMT 2002

In a closed insulated container a liquid is stirred with a paddle to increase the temperature which of the following is true ?

(A) $\Delta E = W \neq 0, q = 0$

(B) $\Delta E = w = q \neq 0$

(C) $\Delta E = 0, W = q \neq 0$

(D) $W = 0, \Delta E = q \neq 0.$

Open solution

Q80. AIPMT 2002

Heat of combustion ΔH^o for $C_{(s)}$, $H_{2(g)}$ and $CH_{4(g)}$ are -94, -68 and -213 kcal/mol, then ΔH^o for $C_{(s)} + 2H_{2(g)} \rightarrow CH_{4(g)}$ is

(A) -17 kcal

(B) -111 kcal

(C) -170 kcal

(D) -85 kcal

Open solution

Q81. AIPMT 2002

Which reaction is not feasible?

(A) $2KI + Br_2 \rightarrow 2KBr + I_2$

(B) $2KBr + I_2 \rightarrow 2KI + Br_2$

(C) $2KBr + Cl_2 \rightarrow 2KCl + Br_2$

(D) $2H_2O + 2F_2 \rightarrow 4HF + O_2$

Open solution

Q82. AIPMT 2002

2 mole of ideal gas at 27^oC temperature is expanded reversibly from 2 lit. to 20 lit. Find entropy change. (R = 2 cal/mol K)

(A) 92.1

(B) 0

(C) 4

(D) 9.2

Open solution

Q83. AIPMT 2001

Change in enthalpy for reaction,

$2H_2O_{2(l)} \rightarrow 2H_2O_{(l)} + O_{2(g)}$

if heat of formation of $H_2O_{2(l)}$ and $H_2O_{(l)}$ are -188 and - 286 kJ/mol respectively, is

(A) $-$ 196 kJ/mol

(B) + 196 kJ/mol

(C) +948 kJ/mol

(D) $-$ 948 kJ/mol

Open solution

Q84. AIPMT 2001

When 1 mol of gas is heated at constant volume temperature is raised from 298 to 308 K. Heat supplied to the gas is 500 J. Then which statement is correct ?

(A) q = w = 500 J, $\Delta E = 0$

(B) q = ΔE = 500 J, w = 0

(C) q = $-$w = 500 J, $\Delta E = 0$

(D) $\Delta E = 0$, q = w = $-$ 500 J

Open solution

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Q85. AIPMT 2001

Enthalpy of $CH_4 + \frac{1}{2} O_2 \rightarrow CH_3OH$ is negative. If enthalpy of combustion of CH_4 and CH_3OH are x and y respectively. Then which relation is correct?

(A) x > y (B) x < y (C) x = y (D) x ≥ y

Open solution

Q86. AIPMT 2001

$PbO_2 \rightarrow PbO$; $\Delta G_{298} < 0$

$SnO_2 \rightarrow SnO$; $\Delta G_{298} > 0$

Most probable oxidation state of Pb and Sn will be

(A) Pb^{4+}, Sn^{4+} (B) Pb^{4+}, Sn^{2+} (C) Pb^{2+}, Sn^{2+} (D) Pb^{2+}, Sn^{4+}

Open solution

Q87. AIPMT 2000

$2Zn + O_2 \rightarrow 2ZnO$; $\Delta G^o = -616$ J

$2Zn + S_2 \rightarrow 2ZnS$; $\Delta G^o = -293$ J

$S_2 + 2O_2 \rightarrow 2SO_2$; $\Delta G^o = -408$ J

ΔG^o for the following reaction

$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ is

(A) − 731 J (B) − 1317 J (C) − 501 J (D) + 731 J

Open solution

Q88. AIPMT 2000

The entropy change in the fusion of one mole of a solid melting at 27°C (latent heat of fusion is 2930 J mol^{-1}) is :

(A) 9.77 J/mol-K (B) 10.77 J/mol-K (C) 9.07 J/mol-K (D) 0.977 J/mol-K

Open solution

Q89. AIPMT 2000

For the reaction,

$C_2H_5OH_{(l)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(l)}$

which one is true

(A) $\Delta H = \Delta E - RT$ (B) $\Delta H = \Delta E + RT$

(C) $\Delta H = \Delta E + 2RT$ (D) $\Delta H = \Delta E - 2RT$

Open solution

Q90. AIPMT 1999

In an endothermic reaction, the value of ΔH is

(A) zero (B) positive (C) negative (D) constant

Open solution

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Q91. AIPMT 1999

In the reaction

$\mathrm{S(s)} + \frac{3}{2}\mathrm{O_2(g)} \longrightarrow \mathrm{SO_3(g)} + 2x\ \mathrm{kcal}$

and $\mathrm{SO_2(g)} + \frac{1}{2}\mathrm{O_2(g)} \longrightarrow \mathrm{SO_3(s)} + y\ \mathrm{kcal}$

the heat of formation of $\mathrm{SO_2}$ is

(A) $(x+y)$ (B) $(x-y)$ (C) $(2x+y)$ (D) $(2x-y)$

Open solution

Q92. AIPMT 1998

Identify the correct statement regarding entropy.

(A) At absolute zero temperature, entropy of a perfectly crystalline substance is taken to be zero

(B) At absolute zero of temperature the entropy of a perfectly crystalline substance is +ve

(C) At absolute zero of temperature the entropy of all crystalline substance is to be zero

(D) At $0^\circ\mathrm{C}$, the entropy of a perfectly crystalline substance is taken to be zero

Open solution

Q93. AIPMT 1998

One mole of an ideal gas at $300\ \mathrm{K}$ is expanded isothermally from an initial volume of $1\ \mathrm{L}$ to $10\ \mathrm{L}$. The ΔE for this process is ($R = 2\ \mathrm{cal\ mol^{-1}\ K^{-1}}$)

(A) $163.7\ \mathrm{cal}$ (B) zero (C) $1381.1\ \mathrm{cal}$ (D) $9\ \mathrm{L\ atm}$

Open solution

Q94. AIPMT 1997

Given that $\mathrm{C + O_2 \rightarrow CO_2}$, $\Delta H^\circ = -x\ \mathrm{kJ}$

$\mathrm{2CO + O_2 \rightarrow 2CO_2}$, $\Delta H^\circ = -y\ \mathrm{kJ}$

The enthalpy of formation of carbon monoxide will be

(A) $\frac{y-2x}{2}$ (B) $2x-y$ (C) $y-2x$ (D) $\frac{2x-y}{2}$

Open solution

Q95. AIPMT 1996

Which of the following is the correct equation?

(A) $\Delta U = \Delta W + \Delta Q$ (B) $\Delta U = \Delta Q - W$ (C) $\Delta W = \Delta U + \Delta Q$ (D) None of these

Open solution

Q96. AIPMT 1995

If enthalpies of formation for $\mathrm{C_2H_{4(g)}}$, $\mathrm{CO_{2(g)}}$ and $\mathrm{H_2O_{(l)}}$ at $25^\circ\mathrm{C}$ and 1 atm pressure are 52, -394 and $-286\ \mathrm{kJ/mol}$ respectively, then enthalpy of combustion of $\mathrm{C_2H_{4(g)}}$ will be

(A) $+141.2\ \mathrm{kJ/mol}$ (B) $+1412\ \mathrm{kJ/mol}$ (C) $-141.2\ \mathrm{kJ/mol}$ (D) $-1412\ \mathrm{kJ/mol}$

Open solution

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Q97. AIPMT 1995

For a reaction to occur spontaneously

(A) ΔH must be negative

(B) ΔS must be negative

(C) $(\Delta H - T\Delta S)$ must be negative

(D) $(\Delta H + T\Delta S)$ must be negative.

Open solution

Q98. AIPMT 1994

During isothermal expansion of an ideal gas, its

(A) internal energy increases

(B) enthalpy decreases

(C) enthalpy remains unaffected

(D) enthalpy reduces to zero.

Open solution

Q99. AIPMT 1994

Following reaction occurring in an automobile

$2C_8H_{18(g)} + 25O_{2(g)} \rightarrow 16CO_{2(g)} + 18H_2O_{(g)}$

The sign of ΔH, ΔS and ΔG would be

(A) $-, +, +$

(B) $+, +, -$

(C) $+, -, +$

(D) $-, +, -$

Open solution

Q100. AIPMT 1991

During isothermal expansion of an ideal gas, its

(A) internal energy increases

(B) enthalpy decreases

(C) enthalpy remains unaffected

(D) enthalpy reduces to zero.

Open solution

Q101. AIPMT 1991

For the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta H = ?$

(A) $\Delta E + 2RT$

(B) $\Delta E - 2RT$

(C) $\Delta H = RT$

(D) $\Delta E - RT$

Open solution

Q102. AIPMT 1990

If ΔH is the change in enthalpy and ΔE, the change in internal energy accompanying a gaseous reaction, then

(A) ΔH is always greater than ΔE

(B) $\Delta H < \Delta E$ only if the number of moles of the products is greater than the number of moles of the reactants

(C) ΔH is always less than ΔE

(D) $\Delta H < \Delta E$ only if the number of moles of products is less than the number of moles of the reactants.

Open solution

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MCQ (More than One Correct Answer) 1 questions

Q103. NEET 2016 Phase 1

The correct thermodynamic conditions for the spontaneous reaction at all temperatures is

(A) $\Delta H < 0$ and $\Delta S > 0$

(B) $\Delta H < 0$ and $\Delta S < 0$

(C) $\Delta H < 0$ and $\Delta S = 0$

(D) $\Delta H > 0$ and $\Delta S < 0$

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **B**	3. **B**	4. **C**	5. **B**
6. **C**	7. **C**	8. **C**	9. **A**	10. **A**
11. **C**	12. **D**	13. **B**	14. **B**	15. **B**
16. **A**	17. **A**	18. **B**	19. **B**	20. **C**
21. **D**	22. **B**	23. **B**	24. **D**	25. **C**
26. **D**	27. **C**	28. **A**	29. **B**	30. **C**
31. **A**	32. **A**	33. **B**	34. **B**	35. **D**
36. **B**	37. **B**	38. **B**	39. **B**	40. **A**
41. **A**	42. **C**	43. **A**	44. **B**	45. **D**
46. **C**	47. **C**	48. **C**	49. **D**	50. **C**
51. **D**	52. **C**	53. **A**	54. **B**	55. **B**
56. **A**	57. **D**	58. **B**	59. **D**	60. **B**
61. **C**	62. **B**	63. **B**	64. **A**	65. **A**
66. **C**	67. **C**	68. **D**	69. **B**	70. **A**
71. **B**	72. **B**	73. **A**	74. **D**	75. **A**
76. **A**	77. **B**	78. **A**	79. **A**	80. **A**
81. **B**	82. **D**	83. **A**	84. **B**	85. **B**
86. **D**	87. **A**	88. **A**	89. **A**	90. **B**
91. **D**	92. **A**	93. **B**	94. **A**	95. **B**
96. **D**	97. **C**	98. **C**	99. **D**	100. **C**
101. **B**	102. **D**			

MCQ (More than One Correct Answer)

103. **A, C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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